MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS3SU
(INCLUDING VARIATIONS LS3SA AND LS3SUS)

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the cable described herein shall consist of this document and the latest issue of MIL-C-24643.

This cable can be supplied in three variations: LS3SU (unarmored), LS3SA (armored) and LS3SUS (double overall shield).

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor, AWG no. 18 (7 by 0.0152 inch), uncoated. Maximum diameter 0.050 inch.

Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.

Third - Cross-linked polyethylene insulation. Nominal thickness 0.023 inch. Colored insulation - one black, one white and one red conductor for each triad.

Fourth - Three conductors (one black, one white and one red) cabled together with a nominal lay of 3.5 inches to form each triad.

Fifth - Binder tape over each triad, at manufacturer's option applied helically with overlap.

Sixth - Braided shield of AWG no. 34 or no. 36 uncoated copper. Braid angle 30 to 35 degrees; minimum coverage of 85 percent.

Seventh - Shield insulation of two polyester tapes type G in accordance with MIL-I-631, sealed. Standard identification code applied by method 2, inner tape. Diameter over completed triad 0.240-inch nominal.

Eighth - The required number of triads (see table I) cabled together with a lay not greater than 24 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with triad no. 1, from the center outward. Fillers as necessary to form a firm, round cable.

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Ninth - Binder tape applied helically with overlap.

Construction variant for type LS3SU

Tenth - Cross-linked polyolefin jacket. (See table I for thickness). Cable surface marking required.

Construction variant for type LS3SA

Tenth - Cross-linked polyolefin jacket. (See table I for thickness).
Eleventh - Braided metal armor and paint.

Construction variant for type LS3SUS

Tenth - Braided shield of AWG no. 34 tin coated copper. Braid angle of 20 to 40 degrees; minimum coverage of 85 percent.
Eleventh - Spirally wrapped 0.002-inch thick polyethylene terephthalate (mylar) separator tape with 25 percent minimum overlap.
Twelfth - Braided shield of AWG no. 34 tin coated copper. Braid angle of 20 to 40 degrees; minimum coverage of 85 percent.
Thirteenth - Binder tape applied helically with overlap.
Fourteenth - Cross-linked polyolefin jacket. (See table I for special thickness). Cable surface marking required.

TABLE I. Details.

Military 1/ part no. M24643/35	Size 2/	Number of triads	Cable jacket thickness inch (nominal)	Overall diameter maximum (inches)					
				LS3SU		LS3SA		LS3SUS	
				min	max	min	max	min	max
-01UN	3	3	0.040	0.650	0.700	0.700	0.750	0.710	0.760
-02UN	7	7	.050	.840	.910	.890	.960	.900	.970
-03UN	10	10	.050	1.100	1.190	1.150	1.240	1.160	1.250
-04UN	14	14	.050	1.200	1.290	1.250	1.340	1.260	1.350
-05UN	19	19	.050	1.340	1.430	1.390	1.480	1.400	1.490
-06UN	24	24	.060	1.580	1.670	1.630	1.720	1.640	1.730
-07UN	30	30	.060	1.680	1.770	1.730	1.820	1.740	1.830
-08UN	37	37	.060	1.840	1.930	1.890	1.980	1.900	1.990
-09UN	44	44	.060	2.060	2.150	2.110	2.200	2.120	2.210

1/ Unarmored. When armor option or double overall shield is required, see MIL-C-24643 for configuration of part number.
2/ Type LS3SU, LS3SA and LS3SUS.

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EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum.	7.06
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor.....	2000
Conductor to shield.....	1000
Shield to shield (pair shields only).....	500
Insulation resistance - shield to shield (LS3SUS only, overall shields only) volt, minimum	6.0
Megohms/1000 feet, minimum.....	1.0
Jacket flaws.....	No failure
Conductor and shield (LS3SUS only) continuity.....	No failure
Group A:	
Visual and dimensional.....	No failure
Capacitance	
Mutual capacitance - pF/ft at 1 kHz, maximum.....	30
Capacitance unbalance - percent, maximum.....	5
Group B:	
Drip - 95 ± 1°C.....	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)..	35
Physicals (unaged)	
Insulation	
Tensile strength - lbf/in^2, minimum.....	1800
Elongation - percent, minimum.....	250
Jacket (cable)	
Tensile strength - lbf/in^2, minimum.....	1300
Elongation - percent, minimum.....	160
Group C:	
Physicals (aged)	
Insulation	
Air oven	
Tensile strength - percent of unaged, minimum.....	80
Elongation - percent of unaged, minimum..	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum.....	60
Elongation - percent of unaged, minimum..	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum.....	50
Elongation - percent, minimum.....	50
Shrinkage.....	No failure
Permanence of printing (jacket) (LS3SU and LS3SUS only) - cycles, minimum.....	250
Heat distortion - percent of unaged, maximum.....	30
Cable filler removability.....	No failure

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EXAMINATION AND TESTS: (Continued)

	Requirements
Shield conformance to material, construction and coverage	No failure
Armor (type LS3SA only) - conformance to material, construction and coverage	No failure
Surface transfer impedance	
Milliohms per meter, maximum	700
EMP response time - ms/500 feet	60

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

Aging and compatibility (cable) (125 ± 5°C)	No failure
Abrasion resistance (jacket/insulation)	No failure
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Flame propagation (cable)	No failure
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	--
Insulation	10
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS:

Size	Feet (nominal)
3 through 19	1000
24 through 44	500

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NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Review activities:
Army - AV, CR, ER
Navy - EC
DLA - IS

User activities:
Army - ME, AR, AL
Navy - CG

Preparing activity:
Navy - SH
(Project 6145-0952)

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